

TIPMAP workflow for TE-SV discovery and TIP-map construction

Variant-centered TE analysis for large pangenomes: extract SV alleles, annotate TE evidence, classify TE-derived alleles, and write a TIP VCF.

Computational motivation

Whole-genome TE annotation for every wheat-scale accession is expensive.

TIPMAP narrows analysis to SV-associated sequences before TE annotation and BLAST-based TE-SV calling.

Inputs

PanPop SV VCF

Merged pangenome SV loci
CHROM, POS, REF, ALT, GT

Pairwise VCF manifest

Genome-to-reference VCFs
genome | vcf | query fasta

Reference FASTA

Reference-side sequence
reference.fa[.fai]

Query FASTA

Optional ALT flank source
ASM_* coordinates

1 Sequence extraction

Generate SV-associated REF and true ALT evidence without full-genome TE annotation.

Parse and normalize VCFs

Infer INS/DEL, SVLEN, END
Expand multiallelic ALT alleles

Genotype-aware filtering

Skip reference-like 0/0
Map ALT1, ALT2, ALT3 carriers

Indexed SV matching

Genome + chromosome + type
Reference POS/END window

Extract sequences

REF from reference genome
ALT from pairwise/query FASTA

Output: sv_ref_alt.fa

all retained REF records and matched true ALT records

2 TE annotation

Annotate only SV-associated sequences and normalize EDTA evidence for downstream classification.

De-duplicate sequences

seqkit rmdup
Reduce redundant SV allele work

Chromosome-level split

Use TIPMAP FASTA headers
Enable parallel EDTA jobs

Run or import EDTA

EDTA.pl --anno 1
or parse existing GFF3

Normalize TE TSV

family, superfamily
coordinates, attributes

Output: te_annotations.tsv

Derived: te_fragments.fa

3 TE-SV identification and TIPMAP output

Classify PanPop alleles as TE-derived, retain TE alleles, and rewrite genotypes into a TIP-map VCF.

Extract PanPop alleles

INS: ALT sequence
DEL: REF sequence

BLASTN comparison

PanPop allele vs TE fragment
outfmt 6

Classify TE alleles

Union coverage ≥ 0.60
Weighted identity ≥ 80
TE-covered bp ≥ 40

Rewrite VCF records

Remove non-TE ALT alleles
Renumeral ALT genotypes

Multiallelic allele handling

Example: if ALT1 and ALT3 are TE-derived but ALT2 is not, 1/1 remains 1/1, 2/2 becomes 0/0, and 3/3 becomes 2/2.

Output: panpop.tip.vcf

Output: panpop.tip_alleles.tsv

Evidence: BLAST TSV

Key outputs and evidence retained by TIPMAP

● Normalized TE annotation TSV preserves EDTA/GFF3 attributes for traceable TE evidence.

● Allele report stores coverage, weighted identity, retained allele number, family/superfamily, and supporting TE annotations.